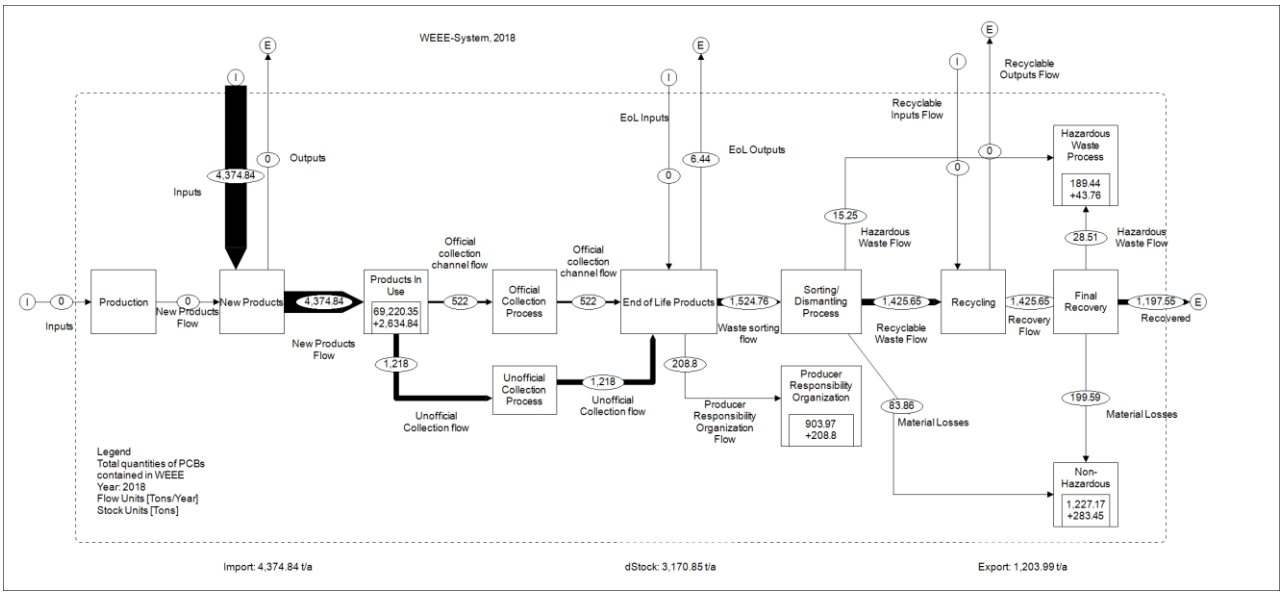
Material Flow Analysis (MFA) for E-Waste Management Using STAN Software: Focus on Printed Circuit Boards (PCBs)

Using **Eurostat datasets (2018)** and **STAN2Web software**, I developed a comprehensive **Material Flow Analysis (MFA)** and mass balance system for Greece. The model was aligned with the EU and National waste management legislation, using the Waste Hierarchy framework (prevention, reuse, recycling, recovery, disposal) as its core structure. The model mapped complex e-waste streams and simulated various waste management scenarios, shifting from conventional landfilling to advanced material recovery and recycling pathways. Additionally, I performed an economic evaluation and environmental impact assessment to quantify the viability of recovering precious metals (Gold, Palladium, Silver) alongside high-volume industrial metals like Copper.



***Note**: The Sankey diagram illustrates the macro-level material flows for the total volume of WEEE, whereas the subsequent tables analyze the specific target metals isolated and studied within the Printed Circuit Boards (PCBs).*

Using this Sankey model, I developed detailed sub-layers for precious metals and copper in Printed Circuit Boards (PCBs). Through this approach, I calculated the emissions from the final recovery of these metals and compared them to the emissions of virgin metal extraction.

Metal	Extraction (t CO_2eq)	Recovered Metals (t CO_2eq)	Avoided Emissions (t CO_2eq)
Gold	15,690.72	25.44	15,665.28
Silver	698.88	17.16	681.72
Palladium	16,856.4	298.2	16,558.2
Copper	1,361,729	269.77	1,361,459.23
Results	**1,394,975**	**610.57**	**1,394,364.43**

According to the results, metal recovery from PCBs represents a strategic pathway toward a sustainable environmental future. Compared to virgin metal extraction, the avoided emissions from recovered metals amount to **1,394,364.43 tCO2eq**. This case study demonstrates that secondary metal recovery effectively mitigates global CO2 emissions while preventing the landscape degradation and ecosystem disruption associated with primary mining. Furthermore, these avoided emissions are **equivalent to the annual greenhouse gas emissions of 303,123 passenger vehicles**. Based on the average carbon sequestration rate of a tree, it is estimated that approximately **55,774,577 trees** would be required to offset an equivalent amount of CO_2 annually.

Metals	Extraction Costs (€)	Recovery Costs (€)	Economic Benefit from recovered metals (€)
Gold	22,732,920	666,693	22,066,227
Silver	980,554	646,058	334,496
Palladium	11,492,172	21,695,458	-10,203,286
Copper	854,583	1,731,560	-876,977
Results	**36,060,229**	**24,739,769**	**11,320,460**

The cost-benefit analysis demonstrates that integrated metal recovery from discarded PCBs is a highly viable economic and environmental alternative to traditional mining. **Gold** and **Silver** serve as the primary financial drivers of the process, with their recovery yields generating massive cost savings that easily offset the financial deficits observed in the more complex extraction of **Palladium** and **Copper**. Ultimately, looking at the system as a whole rather than isolating individual metals reveals a highly profitable cumulative net benefit. This case study proves that while a targeted, stream-specific approach is required, processing e-waste as a collective matrix offers a powerful economic incentive to transition away from resource-intensive primary mining.

Conclusion

In conclusion, this case study highlights that integrated electronic waste management is a vital pillar for both environmental preservation and economic progress. By transitioning from resource-intensive primary mining to secondary metal recovery, we not only secure a highly lucrative financial return driven by precious metals but also achieve massive reductions in global greenhouse gas emissions. Embracing these technologically advanced recycling frameworks is a strategic necessity to prevent ecological degradation, conserve natural landscapes, and protect the planet's morphology for future generations. Ultimately, proving that green practices are financially sustainable provides the ultimate incentive to accelerate the transition toward a true circular economy and long-term environmental sustainability.